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Research Proposal

Improving the Efficiency and Scalability of Training a U-Net Model for Brain Tumour Segmentation under Federated Learning

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Declaration

I, Lily de Melo, hereby declare the contents of this research proposal to be my own work. This proposal is submitted for the degree of Bachelor of Science with Honours in Computer Science at the University of the Witwatersrand. This work has not been submitted to any other university, or for any other degree.

Abstract

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Medical image segmentation is central to brain tumour diagnosis, treatment planning, and disease monitoring, yet training accurate segmentation models requires large and diverse datasets that cannot easily be pooled across hospitals because of privacy, ethical, and regulatory constraints. Federated learning offers a solution by training a shared model from decentralised data, but a standard federated U-Net achieves lower segmentation accuracy and slower convergence than a centrally trained equivalent. It remains unclear which methods and training choices most effectively close this gap, and whether the observed inefficiencies arise from the federated setting itself or from the U-Net architecture. This study proposes an experimental investigation into improving the efficiency and scalability of a federated U-Net baseline for brain tumour segmentation, evaluated on the BraTS 2021 benchmark and conducted on a physically distributed cluster so that communication cost is measured directly rather than simulated. The methodology proceeds in four stages: constructing a non-IID baseline and quantifying the centralised–federated performance gap; systematically analysing the federated training configuration; evaluating targeted methods for handling missing modalities and for reducing communication overhead; and comparing alternative segmentation architectures under identical federated conditions. The intended contribution is to identify the most efficient and scalable configuration for deploying privacy-preserving, clinically viable brain tumour segmentation across heterogeneous hospital networks.

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Chapter 1

Introduction

Medical imaging has become an essential component of modern healthcare by enabling clinicians to visualise internal anatomical structures and abnormalities in a non-invasive manner. One of the most important tasks in medical image analysis is medical image segmentation, which involves partitioning an image into meaningful regions so that anatomical structures or pathological areas can be identified and analysed more precisely [Ronneberger et al. 2015]. In brain tumour analysis, segmentation is particularly valuable because it can help distinguish tumour subregions, supporting diagnosis, treatment planning, and disease monitoring [Menze et al. 2015].

Developing accurate segmentation models often requires large and diverse datasets. In practice, however, such data are usually distributed across multiple hospitals and institutions, and cannot easily be pooled into a single centralised dataset because of privacy, ethical, and regulatory constraints [Sheller et al. 2020]. These limitations make collaborative model training difficult, particularly in medical domains where patient data are highly sensitive.

Federated learning offers a compelling solution to this problem [McMahan et al. 2017; Sheller et al. 2020]. Rather than requiring institutions to share raw patient data, each hospital retains its data locally and contributes only model parameter updates; a central server combines these into a shared global model that is then returned for continued local training [McMahan et al. 2017]. In this way, federated learning supports collaborative model development across multiple institutions without centralising sensitive data [Sheller et al. 2020].

Despite these advantages, federated learning introduces significant challenges in brain tumour segmentation. One major issue is client heterogeneity, which arises

when data across clients are non-independent and identically distributed (non-IID): hospitals differ in their patient populations, scanner hardware, imaging protocols, and disease characteristics, making it difficult for the global model to learn consistently from all clients [Li et al. 2020]. In federated brain imaging, missing modalities, where not all scan types are available for every patient or institution, represent a clinically common and specific form of this heterogeneity [Shi et al. 2026]. A second challenge is communication cost: large model updates must be repeatedly exchanged between clients and the server over many training rounds, and this overhead becomes particularly substantial when architectures such as U-Net are used [McMahan et al. 2017; Chen and Vikalo 2024; Ren et al. 2025].

1.1 Problem Statement

The challenges described above have a direct and measurable impact on federated brain tumour segmentation. Existing work has shown that a standard federated U-Net model achieves lower segmentation performance and converges more slowly than a centrally trained baseline [Stott et al. 2024], demonstrating that adopting the federated framework alone is insufficient. It remains unclear which of these challenges most limits performance, which methods best address them, and whether some inefficiencies are inherent to the federated setting or instead arise from the U-Net architecture itself [Stott et al. 2024; Lemke et al. 2026].

Building on this gap, this study examines methods for handling missing modalities and for reducing communication overhead in a federated U-Net, while also considering how training choices such as mini-batch size and the number of local epochs influence outcomes in federated settings [McMahan et al. 2017; Shi et al. 2026; Chen and Vikalo 2024; Ren et al. 2025].

1.2 Research Questions

1. How does brain tumour segmentation performance compare between a centralised U-Net framework and a standard federated U-Net framework?
2. How do training-configuration choices, namely local epochs and batch size, affect the performance, efficiency, and scalability of a federated U-Net model?
3. Among the methods proposed in the literature, which approaches appear to provide the greatest improvement to federated U-Net brain tumour

segmentation, particularly for handling missing modalities and reducing communication overhead?

4. To what extent do the observed limitations arise from the federated learning setting itself, and to what extent might they be influenced by the U-Net architecture used as the baseline model?

1.3 Research Aims and Objectives

This study aims to identify which approaches are most effective at improving the efficiency and performance of a federated U-Net baseline for brain tumour segmentation, with particular focus on client heterogeneity and communication overhead.

The following objectives support this aim:

1. Compare the segmentation performance of a centralised U-Net with a basic federated U-Net to characterise the performance gap under the federated setting.
2. Analyse how training configuration, specifically local epochs and batch size, can be adjusted to improve the performance, efficiency, and scalability of a federated U-Net model, while acknowledging that client drift may emerge under non-IID conditions.
3. Evaluate methods proposed in the literature for handling missing modalities and reducing communication overhead in federated brain tumour segmentation, and assess which offer the greatest improvement.
4. Assess the extent to which observed limitations arise from the federated learning framework itself versus the U-Net architecture used as the baseline model.

1.4 Outline of the Research Methodology

To address the research questions and aims above, this study adopts an experimental methodology centred on a federated U-Net baseline for brain tumour segmentation evaluated on the full BraTS 2021 benchmark [Baid et al. 2021]. A distinguishing feature is that federated training is conducted on a physically distributed cluster, with each client placed on a separate node and all communication occurring over a real network, so that communication cost is measured directly rather than simulated in memory. The methodology is described in full in Chapter 3.

1.5 Expected Contributions

This study is expected to make the following contributions:

1. A controlled comparative study of centralised and federated U-Net training for brain tumour segmentation, quantifying the performance and convergence gap under a non-IID, multi-channel setting.
2. A reusable distributed federated learning pipeline that runs on a real cluster and measures ground-truth communication cost, namely megabytes transferred and wall-clock transmission time, rather than relying on in-memory simulation.
3. A systematic characterisation of how federated training configuration, namely local epochs and batch size, affects the accuracy, efficiency, and scalability of a federated U-Net, while documenting when client drift becomes a practical concern under non-IID data.
4. A direct comparison of federated learning implementations for missing-modality handling and communication-reduction, evaluated separately and in combination on a common benchmark, dataset partition, and experimental harness, identifying which approach delivers the greatest improvement in the efficiency and scalability of federated U-Net brain tumour segmentation, a comparison the existing literature does not currently provide.
5. Evidence on whether the limitations of federated brain tumour segmentation arise from the federated framework itself or from the U-Net architecture, obtained by benchmarking alternative architectures spanning a range of model capacities under matched federated conditions.

Taken together, these contributions aim to identify the most efficient and scalable configuration for deploying clinically viable brain tumour segmentation across heterogeneous hospital networks.

The remainder of this proposal is organised as follows. Chapter 2 reviews background on federated learning, U-Net, and the BraTS benchmark, then surveys related work across four themes, namely the performance gap between centralised and federated U-Net training, methods for handling client heterogeneity, approaches for reducing communication overhead, and architectural alternatives to U-Net, before synthesising the findings. Chapter 3 sets out the research methodology, including the dataset, the distributed cluster infrastructure, the U-Net model and federated training procedure, the improvement techniques and

architectural comparison, and the evaluation method. Chapter 4 presents the work schedule and timeline. Chapter 5 concludes the proposal.

Chapter 2

Related Work

2.1 Background

As a distributed training paradigm, federated learning was introduced to enable machine learning models to be learned from decentralised data while preserving privacy [McMahan et al. 2017]. Each client performs training locally and sends only model updates to a coordinating server, rather than transferring raw data to a central location. These updates are combined to form a new global model, which is then returned to the clients for continued training [McMahan et al. 2017]. This framework is attractive in settings where data are sensitive or cannot easily be centralised, such as healthcare [Sheller et al. 2020]. A key feature of federated learning is that collaboration occurs through parameter exchange rather than raw data sharing [McMahan et al. 2017; Sheller et al. 2020].

Data heterogeneity across clients, commonly referred to as non-IID data, is a persistent difficulty in federated learning [Li et al. 2020]. In practice, the data stored at different clients often differ in meaningful ways. In a medical context, hospitals may use different scanners, imaging protocols, or serve different patient populations. These differences can make the aggregated global model harder to optimise and may reduce convergence speed or final performance [Li et al. 2020; Hsu et al. 2019]. Such challenges are further amplified when the local task involves a complex model architecture, as is the case in brain tumour segmentation.

The segmentation model central to this study is U-Net, an encoder-decoder architecture for dense prediction that is widely used in medical image segmentation [Ronneberger et al. 2015]. It was designed for tasks where every pixel must be assigned a label rather than only assigning a single class to the whole

image, and its skip connections help preserve fine-grained spatial detail that is important for accurate boundary localisation [Ronneberger et al. 2015]. For this reason, U-Net has become a common baseline in brain tumour segmentation studies [Ronneberger et al. 2015; Stott et al. 2024].

The standard benchmark for evaluating this task is the BraTS dataset [Menze et al. 2015]. BraTS contains multimodal magnetic resonance imaging data, typically including T1, T1-contrast enhanced, T2, and FLAIR scans [Menze et al. 2015]. These modalities provide complementary information about tumour structure and surrounding tissue. Because of its standardised format and widespread use, BraTS provides a useful basis for comparing segmentation models across studies.

Beyond these foundational elements, two specific challenges recur throughout the literature reviewed in this study. The first is missing modalities, a specific form of client heterogeneity in which not all scan types are available for all patients or institutions [Shi et al. 2026]. In multimodal brain imaging, this is a significant challenge because different MRI modalities contribute different information to the segmentation task [Menze et al. 2015]. The second is communication overhead, which refers to the cost of repeatedly transmitting model updates between clients and the central server during federated training [McMahan et al. 2017; Ren et al. 2025]. This overhead can be measured in terms of bandwidth requirements, transmission time, energy consumption, or the number of communication rounds needed for convergence [McMahan et al. 2017; Chen and Vikalo 2024]. One widely studied technique for reducing this cost is quantisation, which compresses model updates by representing parameters with fewer bits, thereby lowering the volume of data that must be exchanged each round [Chen and Vikalo 2024].

2.2 Related Work

Early federated learning work established the FedAvg algorithm and demonstrated that decentralised training could reduce the need to centralise sensitive data [McMahan et al. 2017]. In that framework, clients perform local updates on their own data and a central server aggregates these updates to produce a shared global model. Empirical results showed that increasing local epochs reduced the number of communication rounds required for convergence, and that smaller mini-batches improved learning compared to using the full local dataset [McMahan et al. 2017]. These findings motivate increasing local epochs as a way

to reduce communication cost, but later work showed that under non-IID data, training for many local epochs can cause client updates to drift away from the global objective, eroding this benefit [Li et al. 2020; Karimireddy et al. 2020]. This directly motivates investigating how local epochs and batch size can be configured to improve efficiency in the brain tumour segmentation setting while monitoring the risk of client drift. However, the original work considered general machine learning settings and did not address the full set of challenges that arise in specialised medical imaging applications, including heterogeneous client data, large segmentation architectures, and incomplete multimodal inputs.

Within the context of brain tumour segmentation, Stott et al. [2024] investigated the use of federated learning with a U-Net architecture and compared it with a centrally trained baseline. Their results showed that the federated approach achieved lower segmentation performance and required more time to converge than the centralised model [Stott et al. 2024]. A key factor contributing to this degradation is that the total dataset size was held constant across configurations: as the number of clients increased, each client received a smaller subset of the data, reducing the richness of local training and causing the global model's convergence to depend on which subset each client happened to receive. This study is especially important because it provides a direct baseline for understanding the limitations of federated U-Net training in this domain. It also suggests that training configuration plays an important role in federated performance. Since the study used only one local epoch per federated round, convergence speed and communication cost may both be meaningfully affected by the choice of local epochs and mini-batch size, a possibility supported by the earlier empirical findings of McMahan et al. [2017]. Rather than showing that federated learning is unsuitable for brain tumour segmentation, the study motivates the need to investigate which methods and training choices can improve the efficiency and competitiveness of a federated U-Net baseline.

One important direction in the literature focuses on handling missing modalities as a specific form of client heterogeneity. In real-world multimodal brain imaging data, different institutions or patients may not have access to the same modality combinations, which complicates model training and can reduce segmentation performance. To address this, Shi et al. [2026] proposed FedAMM for federated brain tumour segmentation under arbitrary missing modality settings. Their method uses unimodal prototype distillation during local training and multimodal prototype aggregation at the server to better account for differences in modality availability [Shi et al. 2026]. The reported results show that

FedAMM performs better than baseline methods as modality heterogeneity increases, suggesting that explicitly addressing missing modalities can improve robustness in practical medical settings where complete multimodal data are often unavailable. This makes FedAMM especially relevant to this study, since missing modalities are one of the main sources of heterogeneity considered when improving federated brain tumour segmentation.

Another major direction of research focuses on reducing communication and resource costs in federated learning. This is especially important for large models such as U-Net, where repeated communication rounds may create substantial overhead. Chen and Vikalo [2024] proposed FedMPQ, a mixed-precision quantisation framework that adjusts the bit-width of local models according to client resource constraints. By reducing the precision used to represent model parameters, the method lowers the amount of data that must be transmitted while still maintaining competitive performance [Chen and Vikalo 2024]. This is particularly relevant in settings where the number of communication rounds is large, since smaller updates can reduce the cumulative communication burden over training. A related but distinct approach is presented in TinyMetaFed [Ren et al. 2025], which reduces communication by transmitting only selected global weights with the largest changes rather than sending a full update each round [Ren et al. 2025]. Although TinyMetaFed was developed in a TinyML setting rather than specifically for brain tumour segmentation, it provides an important example of how communication efficiency can be improved either by reducing update precision or by reducing the amount of information transmitted.

A further line of work questions whether the model architecture itself may be part of the limitation. Rather than attempting only to improve training strategy around an existing U-Net model, Lemke et al. [2026] proposed FedNCA, a federated learning framework based on neural cellular automata rather than a standard segmentation architecture such as U-Net. Evaluated on fetal abdominal structures segmentation and clinical intensive care data, rather than brain tumour data specifically, FedNCA achieved comparable segmentation accuracy to federated U-Net and TransUNet baselines while significantly reducing communication overhead and enabling training on low-cost devices [Lemke et al. 2026]. Because FedNCA has not yet been applied to brain tumour segmentation, its findings cannot be transferred directly; however, the principle it demonstrates, that a substantially smaller architecture can match U-Net accuracy in a federated setting at far lower communication cost, raises the question of whether some inefficiencies observed in federated brain tumour segmentation

arise not from the federated framework itself but from the size and design of the U-Net model. For this study, FedNCA therefore motivates considering architectural alternatives as one possible direction for improvement, rather than only adjusting communication strategy or training hyperparameters.

Taken together, these studies show that federated brain tumour segmentation is influenced by several interacting challenges, and that different works address different parts of the problem. Stott et al. [2024] highlight the limitations of a standard federated baseline and show that training settings such as local epochs and batch size may influence convergence behaviour and overall efficiency. FedAMM addresses missing modalities as a form of heterogeneous client data [Shi et al. 2026], while FedMPQ and TinyMetaFed focus on reducing communication and resource burdens through quantisation and selective update transmission respectively [Chen and Vikalo 2024; Ren et al. 2025]. FedNCA further raises the possibility that the underlying segmentation architecture may itself need to change in order to achieve more efficient federated training [Lemke et al. 2026]. The literature therefore points not only to the importance of method design, but also to the importance of training configuration when evaluating a baseline federated U-Net model and determining which changes offer the greatest improvement.

2.3 Conclusion

The reviewed literature shows a consistent pattern but not yet a complete solution. Stott et al. [2024] establish the central problem: a federated U-Net baseline lags behind a centralised counterpart in both segmentation performance and convergence speed. Subsequent work points to three plausible ways to narrow that gap: tuning local training more carefully under non-IID conditions [McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020], explicitly handling missing modalities [Shi et al. 2026], and reducing communication burden through compression or selective transmission [Chen and Vikalo 2024; Ren et al. 2025]. FedNCA further suggests that some of the inefficiency may be architectural rather than purely federated [Lemke et al. 2026].

At the same time, the studies address different parts of the problem in isolation and are rarely evaluated on a common benchmark or experimental harness. The literature therefore provides strong motivation for this study, but not yet a clear recommendation for how best to improve a federated U-Net baseline.

Chapter 3

Research Methodology

This chapter sets out the methodology used to investigate how a federated U-Net baseline for brain tumour segmentation can be made more efficient and competitive with centralised training. The study is designed as a systematic improvement upon the federated U-Net baseline established by Stott et al. [2024], who demonstrated that a standard federated U-Net achieves lower Dice scores and slower convergence than a centrally trained equivalent on BraTS data. While that work identified several contributing factors, it left a number of important issues unaddressed: client data were partitioned in a roughly independent and identically distributed (IID) manner, model input was simplified to a single MRI channel, the training configuration, particularly local epochs and mini-batch size, was fixed and not systematically tuned, no mechanism was introduced for handling missing modalities, no communication-reduction technique was applied, and federation was simulated on a single machine, hiding the true communication cost.

The methodology addresses these gaps in four experimental stages, all run on a distributed cluster so that communication is measured over the network rather than simulated. In the first stage, a non-IID baseline is constructed by introducing Dirichlet-based partitioning of modality availability following Shi et al. [2026], together with a Dirichlet-based skew of client data volume, restoring the full four-channel MRI input, and segmenting the three clinically meaningful tumour subregions; this federated U-Net is then compared directly against a centrally trained U-Net to characterise the performance gap. In the second stage, the training configuration of this baseline is systematically analysed by sweeping the number of local epochs and the mini-batch size, in order to understand how each interacts with convergence behaviour, scalability, and the risk of client drift under non-IID data [McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020]. In the third stage, two families of targeted improvement

methods are evaluated on top of the tuned baseline: a missing-modality handling strategy informed by FedAMM [Shi et al. 2026], and two communication-reduction techniques based on mixed-precision quantisation [Chen and Vikalo 2024] and selective update transmission [Ren et al. 2025]. In the fourth stage, the model-agnosticism of FedAvg is exploited to replace the U-Net backbone with a set of alternative architectures, including lightweight designs such as FedNCA [Lemke et al. 2026], under identical federated conditions, in order to assess whether observed inefficiencies are inherent to the federated framework or attributable to the U-Net architecture itself.

3.1 Datasets

The study uses the full Brain Tumour Segmentation (BraTS) 2021 benchmark [Baid et al. 2021], the standard dataset for evaluating brain tumour segmentation. BraTS 2021 is substantially larger than earlier releases and aggregates data from a greater number of institutions and scanner types, which produces more natural variation across clients and therefore reflects the federated problem being investigated more faithfully than the smaller BraTS 2017 release used by Stott et al. [2024].

All 1,251 labelled BraTS 2021 training volumes are retained in the study pool.

BraTS provides multimodal magnetic resonance imaging (MRI) scans for each patient, comprising four co-registered modalities: native T1, post-contrast T1-weighted (T1ce), T2-weighted (T2), and T2 Fluid-Attenuated Inversion Recovery (FLAIR). Voxel-level annotations identify three nested tumour subregions: the enhancing tumour (ET), the tumour core (TC), and the whole tumour (WT). All four modalities are retained as separate input channels, in contrast to the single-channel simplification used in the prior baseline, and segmentation is evaluated on all three subregions to remain comparable with the broader BraTS literature.

3.2 Data Preparation and Preprocessing

The data pipeline is described in three steps: first the dataset split and federated partition, then preprocessing, and finally patch extraction and augmentation. This ordering separates how the data are assigned from how they are cleaned and from how additional training variation is introduced.

The full BraTS 2021 pool is split once at the patient level into training, validation, and test partitions under a fixed random seed, and these held-out validation and test sets remain identical across every configuration, method, and architecture. The validation set is used for model selection, including comparison of training configurations and selection of the best-performing checkpoint, while the test set is reserved for final reporting only. Within each run, the training partition is then distributed across the K federated clients in deliberately uneven quantities: the proportion of patient volumes assigned to each client is drawn from a symmetric Dirichlet distribution with concentration parameter β [Hsu et al. 2019], so that some clients hold substantially more data than others, mirroring the disparity in patient throughput between large and small hospitals. A small β produces a highly skewed split while a large β approaches an even split; the main experiments fix β at a single moderately skewed value and hold it constant across all sweeps, with an even partition retained as a control reference. Modelling this uneven data volume is a deliberate extension of this study: FedAMM [Shi et al. 2026] distributes data evenly across a fixed set of four clients, whereas this study instead treats client data volume as an explicit, controllable axis of heterogeneity rather than holding it constant. Each patient's complete four-channel volume is treated as an indivisible unit so that all data belonging to a given patient reside on exactly one client, and a minimum per-client floor is imposed so that no client receives too few volumes to train locally.

Heterogeneity between clients is introduced along two clinically motivated axes during this partitioning stage. The first is the quantity of local data, which, as described above, is left deliberately uneven across clients. The second is modality availability: the MRI modalities that accompany each patient's volume vary across samples and clients. This study therefore defines client heterogeneity by data quantity and modality availability rather than by label-distribution skew, since institutions differ in patient throughput and individual patients differ in which scan types were acquired during imaging, rather than in the labels they hold [Shi et al. 2026; Sheller et al. 2020]. Modality availability across samples and clients is controlled by a Dirichlet concentration parameter α, following the scheme introduced by Shi et al. [2026]. When $\alpha = 1$, samples tend to exhibit completely missing modalities, with each sample retaining only one of the four modalities; as α decreases, the setting shifts toward arbitrary missingness, in which samples retain varying modality subsets and modality heterogeneity among clients increases. Experiments sweep $\alpha \in \{0.001, 0.1, 1\}$ to test increasing levels of modality heterogeneity. A full-modality control, in which every

patient volume retains all four modalities and no channels are dropped, is additionally retained as a reference partition. The partition for each α setting is generated once under a single fixed random seed with β held constant, and is then reused across every configuration, method, and architecture compared, so that the data setup is fully reproducible and any observed difference is attributable to the factor under test rather than to a change in the random draw.

Each MRI volume is then preprocessed identically across the centralised and federated pipelines so that any performance differences can be attributed to the training paradigm rather than to differences in data preparation. Volumes are resampled to a common voxel spacing, skull-stripped using the masks provided with the BraTS release, and intensity-normalised on a per-modality, per-patient basis using z-score normalisation computed over non-zero voxels. The four modalities are then stacked to produce a four-channel input tensor.

To make training computationally tractable while preserving the dense-prediction nature of the task, fixed-size three-dimensional patches are extracted from each volume during training, with patch sampling weighted toward tumour-containing regions to address foreground–background class imbalance. Standard on-the-fly augmentations, including random flips along anatomical axes, small random rotations, and intensity perturbations, are applied during local client training only. At inference time, full volumes are processed using a sliding-window strategy with patch overlap to avoid boundary artefacts.

Modality availability is enforced per patient volume on the node where that volume resides: when a sample is assigned a given modality subset, the absent modality channels are simply not present in that volume's local storage. Because missingness is assigned per sample rather than per client, a single client typically holds samples spanning many different modality subsets, so the modalities present vary both within and across clients. A binary modality-availability descriptor is published alongside each patient volume, recording which of the four modalities it contains, so that the federated training procedure and the FedAMM-inspired prototype mechanism described in Section 3.4.4 can route around the channels missing from each individual sample.

3.3 Distributed Federated Learning Infrastructure

A central methodological choice of this study is to run federated training on a physically distributed cluster rather than to simulate federation in memory on a single machine. The experimental setup provides up to eight client nodes, each running exactly one client, together with one further node designated as the central server responsible for aggregating model updates. To assess scalability across different federation sizes, the number of clients K is swept over $K \in \{3, 4, 5, 6, 8\}$ within the limits of the available cluster nodes; in any given run, K nodes are active, and all K clients participate in every communication round, so the participating subset that the server broadcasts to in Section 3.4.2 is always the complete active client set, corresponding to a client fraction of $C = 1$. All communication between clients and the server takes place over the cluster network, so bandwidth consumption and transmission time per round are measured directly rather than inferred from an in-memory simulation.

Each client trains only on the local subset of the BraTS data stored on its node, and the data partition assigned to a client never leaves that node, mirroring the privacy-preservation property that motivates federated learning in the medical domain [Sheller et al. 2020]. The server runs the FedAvg aggregation logic described in Section 3.4.2 and broadcasts the resulting global parameters back to clients at the start of each round. A synchronisation barrier at the server ensures that aggregation begins only once all participating clients have completed their local training for the round, so that round-level measurements are well defined and comparable across configurations. Because all client nodes are identical, any measured difference in convergence behaviour or final segmentation quality across configurations is attributable solely to the training or compression scheme under test, not to differences in compute speed or local hardware capability.

3.4 Models and Algorithms

3.4.1 Segmentation Backbone

The segmentation model used throughout this study is a three-dimensional U-Net [Ronneberger et al. 2015], retained as the architectural backbone to keep the comparison with Stott et al. [2024] direct and to isolate the effect of federated training choices from the effect of architectural changes. U-Net is appropriate for this problem for two reasons. First, brain tumour segmentation

is a dense prediction task that requires a label for every voxel, and U-Net's encoder–decoder design with skip connections preserves the fine-grained spatial detail needed for accurate boundary localisation between tumour subregions [Ronneberger et al. 2015]. Second, U-Net remains the dominant baseline in the federated brain tumour segmentation literature [Stott et al. 2024; Shi et al. 2026], which makes results reported in this study directly comparable with prior and concurrent work.

The network follows the standard symmetric encoder–decoder topology with four resolution levels. Each encoder block applies two padded $3 \times 3 \times 3$ convolutions, instance normalisation, and a non-linear activation, followed by a strided convolution that halves spatial resolution. Each decoder block performs transposed-convolution upsampling, concatenates the corresponding skip connection from the encoder, and applies two further $3 \times 3 \times 3$ convolutional layers. A final $1 \times 1 \times 1$ convolution projects features to the number of tumour classes. The input layer accepts four channels corresponding to the BraTS modalities, restoring the multi-channel input that was simplified in the prior baseline.

3.4.2 Federated Training Procedure

Federated training follows the FedAvg framework [McMahan et al. 2017]. Let K denote the number of clients, with client k holding a local dataset $\mathcal{D}_k$ of size n_k, and $n = \sum_{k=1}^{K} n_k$. At communication round t, the server broadcasts the current global parameters w_t to a participating subset of clients. Each selected client k initialises its local model with w_t and performs E local epochs of mini-batch stochastic gradient descent with batch size B on $\mathcal{D}_k$, producing updated parameters w_{t+1}^k. The server then aggregates these updates by a sample-size weighted average:

$$w_{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} \, w_{t+1}^k. \tag{3.1}$$

Training proceeds round by round until validation Dice shows no further improvement; the resulting number of communication rounds is denoted by T.

3.4.3 Training Configuration Analysis

Because Stott et al. [2024] used a single fixed configuration of one local epoch per round, the second stage of the methodology systematically varies the FedAvg hyperparameters identified by McMahan et al. [2017] as the primary controls on convergence and communication cost: the number of local epochs E and

the mini-batch size B. Because every active client participates in every communication round (Section 3.3), the client fraction is held fixed at $C = 1$, and the analysis in this subsection is conducted at a selected client count K chosen from the scalability sweep described in Section 3.3. Anchored in the values used by McMahan et al. [2017], the initial grid takes $E \in \{1, 5, 10, 20\}$ and $B \in \{4, 8, 16\}$, the latter sized for volumetric MRI patches.

Because increased local work under non-IID data can cause client updates to drift from the global objective [Li et al. 2020; Karimireddy et al. 2020], this stage records how training stability changes as E and B vary, rather than introducing separate drift-control algorithms. The aim of this stage is therefore not only to find a strong configuration but also to characterise the point at which additional local work begins to harm rather than help convergence due to client drift. Candidate values of E and B are compared on the held-out validation set, and the best-performing checkpoint for each run is selected from validation performance. Convergence behaviour is measured both in rounds-to-target-Dice and in cumulative communicated bytes, so that efficiency and scalability can be discussed jointly with accuracy.

3.4.4 Improvement Strategies

Three improvement strategies are implemented and evaluated: a missing-modality-enabled system, a communication-efficient system that itself combines two complementary mechanisms, and a combined system that enables both. Each is compared against the same tuned baseline so that its contribution can be isolated.

Missing-modality-enabled federated learning. The first configuration targets the heterogeneity that arises when individual patient volumes are accompanied by only a subset of the four MRI modalities. It builds on the prototype-based strategy of FedAMM [Shi et al. 2026]. The naive missing-modality baseline is a standard four-channel federated U-Net. In this baseline, absent modality channels are zero-filled at the client data loader, which requires no architectural change. The FedAMM-inspired prototype mechanism is evaluated against this zero-fill baseline under identical data partitions, α values, and training configuration.

The prototype mechanism's implementation has three parts. First, the 3D U-Net encoder is restructured so that each modality is processed by its own lightweight encoder stem. The per-modality feature maps are then fused by a

learned weighted sum before entering the shared decoder. Every client instantiates all four encoder stems, since missingness is assigned per sample and a client generally holds samples that, taken together, cover every modality. For each training sample, only the stems corresponding to the modalities present in that sample are activated, and a given stem is updated only by the samples that contain its modality. This matches the per-sample enforcement of missing modalities described in the data preparation and preprocessing section.

At the end of each local training round, each client computes prototype vectors from the deepest encoder features of its model. A prototype represents the average bottleneck feature vector for a specific MRI modality and tumour subregion. These prototypes act as compact summaries of how each modality represents each tumour class. The server collects the prototypes into a global prototype bank. During later local training, each client adds a distillation loss that encourages its own voxel-level bottleneck features to be close to the relevant global prototypes. This allows the representations learned from samples with few available modalities to benefit from the richer representations learned from samples with more complete modality sets, both within and across clients, without sharing raw patient data.

Third, the server aggregates the prototypes alongside the FedAvg parameter average. For each modality, prototypes are averaged across clients in proportion to the number of samples at each client that contain that modality, so that a client contributes to a modality's prototype only to the extent that it holds samples providing it. The resulting global multimodal prototype bank is broadcast together with the global parameters w_t. The per-sample modality-availability descriptors published during preprocessing tell the server how many samples of each modality each client holds, and likewise let each client activate only the encoder stems relevant to each sample.

Communication-efficient federated learning. The second configuration reduces the data-transfer cost incurred by transmitting full model weights every round, using two complementary mechanisms.

The first is mixed-precision quantisation of the transmitted updates, following the principle of FedMPQ [Chen and Vikalo 2024]: a quantisation operator maps a float32 tensor to a b-bit integer representation using a per-channel scale factor $s = \max(|w|)/(2^{b-1} - 1)$, so that each parameter w is transmitted as the integer $q = \text{round}(w/s)$ together with the scale s and reconstructed as $\hat{w} = s\,q$; the server dequantises every update to float32 before the weighted

FedAvg aggregation, and the residual $w - \hat{w}$ is retained locally and added to the following round's update so that quantisation error does not accumulate as bias. Although the experimental nodes are homogeneous, real-world federations comprise institutions with heterogeneous compute and bandwidth constraints [Chen and Vikalo 2024]. To simulate this, each client is assigned a synthetic resource profile that determines its bit-width budget b_k. In the homogeneous-precision experiments, all clients share a single bit-width $b \in \{2, 4, 8\}$ so that the effect of quantisation precision on model quality and communication cost can be isolated independently of any other variable. In the heterogeneous-precision experiments, clients are assigned bit-widths from $\{2, 4, 8\}$ in fixed proportions (e.g., one third of clients at each level), following the experimental setup of Chen and Vikalo [2024]. Because the underlying nodes are identical, any difference in convergence behaviour or final Dice score between these configurations is attributable solely to the quantisation scheme.

The second mechanism is selective update transmission, inspired by TinyMetaFed [Ren et al. 2025]: after local training the client forms the update $\Delta_k = w_{t+1}^k - w_t$, ranks its entries by absolute magnitude, and transmits only the top-ρ fraction as a sparse set of (index, value) pairs, with the server treating every unsent entry as zero change; the untransmitted residual is accumulated locally so that parameters which change slowly but consistently are eventually sent in a later round. The transmitted fraction is varied over $\rho \in \{0.01, 0.1, 0.25, 0.5\}$, where $\rho = 0.01$ transmits only the 1% of parameters with the largest absolute changes and $\rho = 0.5$ transmits half; TinyMetaFed [Ren et al. 2025] experiments with $P \in 10\%$–80%, while Lemke et al. [2026] evaluate top-1% and top-25% in their federated compression experiments, and the range chosen for this study spans aggressive to moderate sparsity. The uncompressed baseline transmits the full float32 parameter vector each round with no sparsification (i.e., $b = 32$, $\rho = 1.0$), corresponding to the tuned federated U-Net from the configuration analysis, and all compression results are reported relative to this baseline.

The two mechanisms are evaluated in three groups. First, quantisation-only: each bit-width $b \in \{2, 4, 8\}$ is applied with $\rho = 1.0$, so that all parameters are transmitted. Second, selective-only: each sparsity level $\rho \in \{0.01, 0.1, 0.25, 0.5\}$ is applied at full float32 precision. Third, the combined configuration applies quantisation to the values of the entries selected by the top-ρ mask, evaluated at a subset of the (b, ρ) grid identified from the first two groups as Pareto-optimal on the Dice-versus-bytes frontier. This three-group design isolates the

contribution of reduced precision from the contribution of reduced volume before testing their interaction.

Combined configuration. The third configuration evaluates a single system in which the missing-modality mechanism and the communication-reduction mechanisms are enabled together, so that quantisation and selective transmission act on both the model parameters and the modality prototypes exchanged each round. This configuration is run as its own experiment rather than as an extension of the communication-efficient setup, allowing the interaction between the two ideas to be assessed directly.

It tests two things that neither component reveals alone: whether the communication savings measured for the plain U-Net still hold once the additional prototype payload is introduced, and whether compressing the prototypes degrades the prototype-based knowledge transfer that the missing-modality method depends on.

Because all training takes place on the distributed cluster, every configuration reports cumulative megabytes transmitted and wall-clock transmission time per round measured directly from the network, showing each strategy's practical benefit and any accompanying change in convergence speed.

3.4.5 Architectural Comparison

The final stage investigates whether the limitations observed in the earlier stages are inherent to the federated framework or attributable to the U-Net architecture. Because the FedAvg algorithm is model-agnostic and imposes no architectural constraints [McMahan et al. 2017], the segmentation backbone can be exchanged while every other element of the pipeline is held fixed. Each candidate architecture is therefore dropped into the unchanged FedAvg pipeline, with its parameters taking the place of the global parameters w and broadcast, locally trained, and aggregated exactly as in the FedAvg procedure already described. The data partition, the Dirichlet concentrations α and β, the client count K, the client-to-node placement, the communication-round budget, and the random-seed settings are held fixed across all architectures, so that the data and federation conditions are identical. The local-iteration count E and the batch size B, however, cannot simply be transferred: a value that is optimal for U-Net is not guaranteed to be optimal for a transformer or a neural-cellular-automata model, and reusing U-Net's settings unchanged would systematically advantage U-Net. Each alternative architecture therefore receives a brief, coarse

re-tuning of E and B around the baseline values before its comparison run, so that every architecture is evaluated near its own operating point. This light per-architecture tuning is a deliberate trade-off: it avoids an unfair comparison while keeping the additional compute small, at the cost of E and B no longer being held literally identical across architectures.

The principal alternative is FedNCA [Lemke et al. 2026], a federated segmentation model based on neural cellular automata. The volume is represented as a grid of cells, in which each cell carries a state vector comprising the four input modality intensities together with a number of additional hidden channels. A single small update network, consisting of a few 3D convolutional layers and on the order of a few thousand parameters, defines a local update rule that is applied identically to every cell. This rule is applied iteratively for a fixed number of steps, after which the segmentation logits are read from designated output channels of the final cell state. Because the same compact rule is reused at every step and every spatial location, the parameter count is orders of magnitude smaller than that of U-Net, which is the source of FedNCA's communication advantage.

A single alternative, however, cannot separate the effect of model size from the effect of model design. Two further architectures are therefore placed alongside FedNCA, each likewise integrated without modification into the federated pipeline, chosen so that the set spans as wide a range of model capacity as possible:

- **A lightweight depthwise-separable U-Net.** A U-Net variant in which the standard $3 \times 3 \times 3$ convolutions are replaced by depthwise-separable convolutions and the channel widths are reduced. This retains the conventional encoder–decoder design but cuts the parameter count substantially, testing whether shrinking U-Net within its own paradigm recovers FedNCA's communication efficiency without adopting the cellular-automata formulation.
- **UNETR** [Hatamizadeh et al. 2022]. A transformer-based 3D segmentation network with a substantially larger parameter count than U-Net. As the heavyweight end of the spectrum it tests whether additional modelling capacity narrows the centralised–federated gap or instead inflates communication cost without a commensurate accuracy gain.

Together with the baseline U-Net itself, these architectures span from a few thousand parameters (FedNCA), through a shrunk depthwise-separable U-Net

and the standard U-Net, to a heavyweight transformer, placing model capacity on a single comparison axis. Evaluating them against the tuned federated U-Net on identical data and partitions, with only the light per-architecture tuning of E and B described above, places the centralised–federated performance gap and the communication cost on a common axis of parameter count and design. If a substantially lighter model matches the Dice score of the tuned federated U-Net at far lower communication cost, the observed inefficiency was architectural; if accuracy instead tracks capacity regardless of architecture family, the limitation lies with the federated framework itself, directly addressing the open question raised by Lemke et al. [2026].

Table 3.1 consolidates the experimental grid that these four stages define. Each parameter is swept while the remaining parameters are held at their baseline values, so that any observed change in segmentation quality or communication cost is attributable to the factor under test.

TABLE 3.1: Summary of the experimental grid defined by the four methodology stages. Each parameter is swept while the others are held at their baseline values, so that any observed difference is attributable to the factor under test.

Parameter	Symbol	Values explored	Stage
Client count	K	3, 4, 5, 6, 8	1
Local epochs	E	1, 5, 10, 20	2
Mini-batch size	B	4, 8, 16	2
Client fraction	C	1 (fixed)	1–4
Data-quantity skew	β	Moderately skewed (fixed); even partition as control	1–4
Modality-availability skew	α	0.001, 0.1, 1	1–3
Quantisation bit-width, homogeneous	b	2, 4, 8	3
Quantisation bit-width, heterogeneous	b_k	Mix of {2, 4, 8} in fixed proportions	3
Selective-transmission fraction	ρ	0.01, 0.1, 0.25, 0.5	3
Uncompressed baseline	(b, ρ)	(32, 1.0)	3
Segmentation architecture	—	U-Net; depthwise-separable U-Net; UNETR; FedNCA	4

3.5 Evaluation Method

The evaluation protocol is designed to be directly comparable both to Stott et al. [2024] and to the broader BraTS literature, and to capture the joint trade-off between segmentation quality and federated efficiency that motivates the study. Validation data are used for model selection during development, whereas all final performance figures reported in the study are taken from the held-out test set.

Segmentation quality is reported using the metrics standard to the BraTS benchmark [Menze et al. 2015]. The primary metric is the Dice similarity coefficient reported separately for each of the three nested tumour subregions (enhancing tumour, tumour core, whole tumour) and averaged across the test cohort, following the standard BraTS protocol and enabling direct comparison with the broader medical imaging literature. The Dice coefficient between a predicted segmentation P and ground truth G is

$$\mathrm{Dice}(P, G) = \frac{2\,|P \cap G|}{|P| + |G|}. \tag{3.2}$$

The 95th-percentile Hausdorff distance (HD_{95}) is reported alongside Dice as a complementary boundary-based metric, since Dice alone can mask localised boundary errors that are clinically significant.

Sensitivity, also termed recall, is additionally reported per subregion as a complementary directional metric:

$$\mathrm{Sensitivity} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}, \tag{3.3}$$

where TP and FN are the voxel-level counts of true positives and false negatives. Because Dice is a symmetric overlap measure, it cannot distinguish under-segmentation from over-segmentation; Sensitivity isolates the proportion of true tumour tissue that is correctly identified, so a low value specifically signals missed tumour, which is clinically the more serious error. Sensitivity is also among the metrics defined for the original BraTS benchmark [Menze et al. 2015], supporting comparison with the broader literature.

For the missing-modality experiments, the held-out test set is evaluated under two conditions. First, a full four-channel evaluation confirms that the prototype-trained model does not regress on complete inputs compared with the zero-fill

baseline. Second, modalities are withheld from test volumes to match the training distribution at each α value, so that robustness is measured consistently. Results are additionally reported for each of the 15 non-empty modality combinations, following the evaluation protocol of Shi et al. [2026].

Convergence efficiency is tracked as the communication round at which the global model first attains 90% and 95% of its final Dice score, giving a concrete efficiency measure beyond final accuracy and a direct quantification of the slower-convergence finding reported by Stott et al. [2024].

Because federated training runs on a physically distributed cluster, communication cost is measured directly from the network. The transmission time per round and the cumulative megabytes transferred are recorded for each configuration. These measurements jointly capture the effect of update size (as reduced by quantisation or selective transmission) and the number of rounds required for convergence. Communication-reduction results are reported as Pareto-optimal configurations on the Dice score versus cumulative megabytes transmitted frontier: a configuration dominates another if it achieves equal or higher Dice at equal or lower communication cost. This avoids an arbitrary accuracy threshold and lets the reader assess the trade-off directly.

Per-client performance is recorded throughout training in addition to the aggregate global score, in order to quantify the struggling-client problem in which some nodes consistently learn less effectively due to the composition of their local data. The variance of per-client Dice across clients, and its evolution across rounds, are reported separately so that gains from missing-modality handling and from drift-related effects of training configuration can be distinguished from improvements in average accuracy.

All experiments are run under a single fixed random seed, so that client sampling, partitioning, and weight initialisation are identical across every configuration, method, and architecture; any observed difference therefore reflects the factor under test rather than a change in the data setup or initialisation.

3.6 Scope and Limitations

Three limitations bound the conclusions this study can support and are stated here so that its results are interpreted appropriately.

First, all experiments use a single benchmark, BraTS 2021. The study therefore measures relative behaviour across configurations, methods, and architectures

under controlled conditions, and does not establish that the findings transfer to other datasets or to genuinely unseen institutions; external cross-dataset validation is left to future work.

Second, the missing-modality and alternative-architecture methods evaluated here, namely FedAMM [Shi et al. 2026] and FedNCA [Lemke et al. 2026], are re-implementations guided by their original publications rather than the authors' released systems. The study consequently tests the principles these methods embody, prototype-based modality transfer and a compact cellular-automata backbone, under a common harness; any divergence from the originally reported results cannot be attributed unambiguously to the method rather than to the re-implementation, and the comparison should be read as a like-for-like evaluation within this harness rather than a reproduction of the original papers.

Third, to keep the experimental grid within compute budget, each experiment is run once under a single fixed random seed rather than repeated across several. Reported scores therefore carry no estimate of seed-to-seed variability, and a small difference between two configurations, methods, or architectures cannot be confidently distinguished from the variation a different seed would have produced. The comparisons drawn in this study are accordingly treated as strongest where the differences observed are large and consistent across related experiments, and a narrow margin between two settings is reported as inconclusive rather than as a definitive ranking.

3.7 Conclusion

The methodology is designed to isolate which changes genuinely improve a federated U-Net baseline. A stronger non-IID baseline is first constructed on BraTS 2021; the training configuration is then tuned; missing-modality and communication-reduction methods are evaluated on top of that tuned system; and finally an architectural swap is tested to determine whether it changes the picture. Throughout, segmentation quality, convergence behaviour, per-client variation, and network-measured communication cost are reported against centralised and baseline federated references so that each intervention can be judged on the same footing.

Chapter 4

Work Schedule

The work plan covers the five-month period from July to the end of November 2026. It separates implementation from experimentation: July is dedicated to provisioning the distributed cluster environment and implementing every component of the pipeline, after which the experiments for the four research aims are run over the following two months, with August covering Aims 1 and 2 and September covering Aims 3 and 4. October is reserved for writing the research report and November for clean-ups and final submission. Cluster setup is folded into the start of July, since the distributed federated learning infrastructure is a prerequisite for every subsequent experiment.

4.1 Milestones

1. **July - Implementation.** Provision the distributed cluster environment, configuring one node as the central server and the remaining nodes as clients, and implement the FedAvg communication protocol over the cluster network with a synchronisation barrier at the server. Prepare the full BraTS 2021 dataset and put the preprocessing pipeline in place (resampling, skull-stripping, z-score normalisation, four-channel stacking, Dirichlet-based modality-distribution partitioning), ensuring that each client is assigned at least m patient volumes so that local training remains data-rich. With the infrastructure in place, implement every model and method required by the later aims: the centralised and federated U-Net used in Aims 1 and 2, the FedAMM-inspired missing-modality system and the FedMPQ- and TinyMetaFed-style communication-efficient system used in Aim 3 [Shi et al. 2026; Chen and Vikalo 2024; Ren et al.

2025], and the model-agnostic architecture-swap harness that lets the U-Net backbone be exchanged within the unchanged FedAvg pipeline for Aim 4 [Lemke et al. 2026].

2. **August - Aims 1 and 2.** Carry out Aim 1 by running both the centralised U-Net and the federated U-Net on the multi-channel non-IID partition, reporting per-subregion Dice and HD_{95} and quantifying the performance gap. Then carry out Aim 2 by first sweeping the client count K across the available federation sizes and then, at a selected K, sweeping the number of local epochs and the mini-batch size across the three modality-heterogeneity settings $\alpha \in \{0.001, 0.1, 1\}$, and identify the configuration that gives the best observed accuracy–communication trade-off for use in the later aims, while noting when client drift begins to degrade convergence.
3. **September - Aims 3 and 4.** Carry out Aim 3 by running the targeted improvement experiments on the tuned baseline: the FedAMM-inspired missing-modality handling with node-level enforcement of missing modalities [Shi et al. 2026], the FedMPQ-style mixed-precision quantisation of transmitted updates [Chen and Vikalo 2024], and the TinyMetaFed-style selective update transmission [Ren et al. 2025], evaluating each technique independently and the two communication-reduction techniques in combination, and recording per-client Dice variation and ground-truth network cost throughout. Then carry out Aim 4 by running the architectural comparison, swapping the FedNCA backbone [Lemke et al. 2026] and the other alternative architectures into the unchanged FedAvg pipeline under otherwise identical federated conditions and comparing against the tuned federated U-Net on the same metrics to determine whether observed limitations are architectural or inherent to federated learning.
4. **October - write-up.** Consolidate the results across all four aims, produce the final figures and tables, and draft the full research report. End of October: complete first draft submitted to the supervisor for feedback.
5. **November - clean-ups and submission.** Address supervisor feedback, perform any minor re-runs needed for missing data points or clarifying ablations, polish the writing, and prepare the final submission package. End of November: final submission.

4.2 Timeline

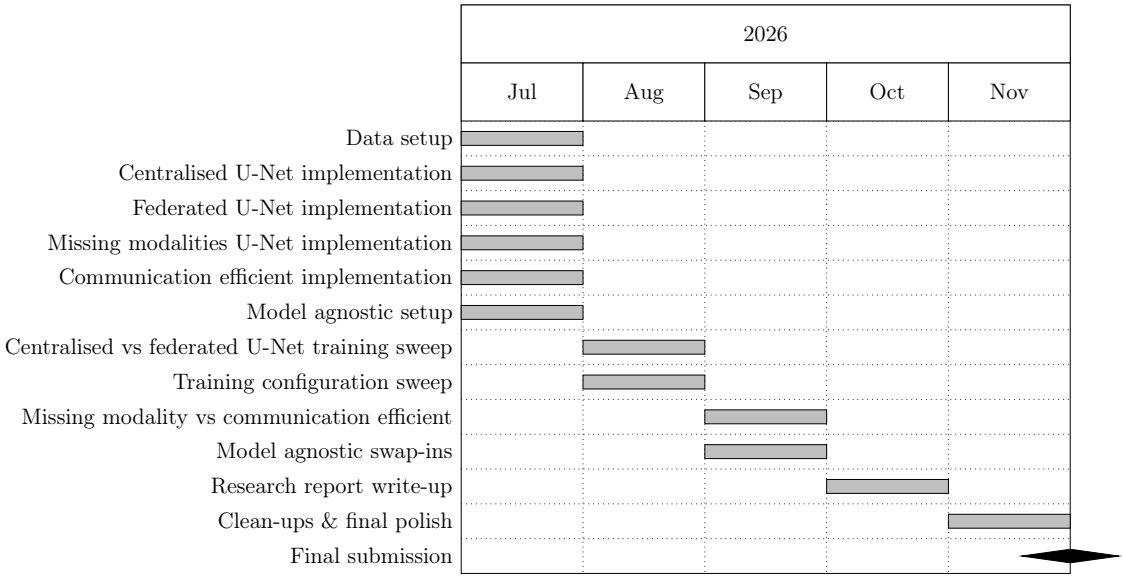


FIGURE 4.1: Five-month project schedule from July to November 2026. July covers cluster setup and the implementation of every pipeline component; the experiments then run over the next two months (August: Aims 1–2; September: Aims 3–4); October is for write-up and November for clean-ups and submission.

4.3 Risks and Contingencies

The most significant risk to the timeline is ensuring that the distributed federated training pipeline runs reliably in a real cluster environment. Unlike the in-memory simulation used by prior work, deployment on physically separate nodes introduces genuine inter-node communication, job scheduling, and resource-contention behaviour that cannot be fully anticipated in advance, and every later experiment depends on this pipeline working. Difficulties in cluster provisioning or in bringing up the cluster-level communication protocol could therefore delay the start of all subsequent aims. To mitigate this, cluster smoke tests on a small subset of BraTS 2021 will begin in the first week of July rather than later, so that integration issues surface early. If the resulting delay compresses the time available for the August experiments, the Aim 2 training-configuration sweep will be pruned first by identifying the more influential factor between local epochs and batch size through a coarse scan and concentrating the remaining time on that axis. Client drift under non-IID data is acknowledged as a further methodological risk: if larger local-epoch settings prove unstable, the final tuned configuration will favour the most stable operating point rather

than the most aggressive one. The Aim 4 architectural comparison in September is the most easily descoped: if the FedNCA backbone is not stable in time, it can be reported as a single comparison run rather than a full sweep without affecting the conclusions of Aims 1–3.

Chapter 5

Conclusion

The intended contribution of this study is to identify which approach will best improve the efficiency and scalability of federated brain tumour segmentation, by systematically evaluating four core dimensions on a real distributed cluster rather than in simulation. *Training configuration* is benchmarked across local epochs and batch size to establish strong settings, convergence behaviour, and scalability for a federated U-Net baseline, while documenting when client drift becomes a practical limitation under non-IID training. *Missing modalities* are adopted as the central axis of client heterogeneity, alongside deliberately uneven client data volumes, with modality availability distributed across clients by a Dirichlet-based scheme [Shi et al. 2026], and are addressed by evaluating a FedAMM-style modality-prototype mechanism against a naive zero-fill baseline for robust performance when clients lack scan types. *Communication efficiency* is evaluated through compression techniques, namely mixed-precision quantisation [Chen and Vikalo 2024] and selective update transmission [Ren et al. 2025], that aim to reduce transfer costs while preserving segmentation quality. *Architecture selection* is examined by benchmarking FedNCA [Lemke et al. 2026] and further architectures spanning a range of model capacities against U-Net under matched federated conditions, in order to attribute observed limitations either to the federated framework or to the U-Net architecture itself.

Together, these evaluations are intended to yield an evidence-based recommendation for deploying privacy-preserving brain tumour segmentation efficiently and at scale across heterogeneous hospital networks.

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